eBook Subscriber Intelligence

Who stays, who leaves, and what to do about it — a subscriber engagement analysis with a live Streamlit dashboard

Krisha Kadam · Industrial Research Project · October 2026

5,455 subscribers scored · recency + logins + session minutes · Python · pandas · Streamlit

What this project is

One project, one question — what makes an eBook subscriber stay?

The question

Which subscribers are still reading, which have gone quiet, and which are already gone — and what should the business do with each group?

The data

Two files, one project: the customer file (16,404 customers, 33.2% subscribed) and the engagement file (5,455 subscribers with real reading behaviour).

The answer

A 0-100 engagement score per subscriber, an Active / Inactive / Churned status, and a recommended strategy for each group.

Most subscribers are staying

55.3% of classified subscribers are Active (logged in within 30 days). 28.0% are Inactive — quiet for 31-90 days and still win-backable. 16.6% are Churned — silent for more than 90 days.

Every number in this deck comes from the dataset — nothing is simulated or invented.

How the analysis works

Five steps — every rule approved in writing before it was applied

1 Clean

Drop 45 duplicate rows, collapse 23 genre spellings into 8 genres, flag 144 median-filled rows. Recency is never invented.

2 Explore

Charts and correlations first — what the data shows decides what goes into the score, not the other way round.

3 Score

100 x (0.40 x inverted recency + 0.35 x logins + 0.25 x session minutes). Books excluded: r = 0.88 with logins.

4 Classify

Active <= 30 days - Inactive 31-90 - Churned > 90. 33 rows with missing recency stay Unclassified.

5 Act

Three strategy cards — Retain, Win back before day 90, Win-back — each tied to a finding and a literature theme.

Weights and thresholds were proposed and approved before use - every chart carries a takeaway caption - associations are reported as associations, never as causes.

The engagement score

One number per subscriber — and every ingredient visible

Score = 100 x (0.40 x inverted recency + 0.35 x logins + 0.25 x session minutes)

Recency (days since last login)
inverted - fewer days scores higher



Average logins per month
how often they come back



Average session minutes
how long they stay



Why books is not in the score

Books accessed correlates r = 0.88 with logins - both measure usage frequency. Counting both would weight the same behaviour twice, so books stays descriptive (and becomes a useful independent check later).

How the score becomes a tier

Terciles - the data's own one-third and two-third marks: Low <= 40.80 - Medium <= 51.25 - High above. Equal thirds, no invented round-number cutoffs.

The honest caveat

Three of four inputs reappear in the behaviour charts, so part of the tier separation is by construction - the books panel is the independent test.

Who is Active, Inactive or Churned

Recency thresholds — approved in writing before being applied

Active	**3,001**	**55.3%**	Logged in within the last 30 days, at least 1 login/month
Inactive	**1,520**	**28.0%**	Quiet for 31-90 days (or under 30 days with zero logins)
Churned	**901**	**16.6%**	No activity for more than 90 days — median 152 days
Unclassified	**33**	**—**	Recency missing or invalid — never guessed, shown separately

Status is recency-based by definition — that is why the behavioural comparisons in this deck are honest: the chart is not simply repeating the rule.

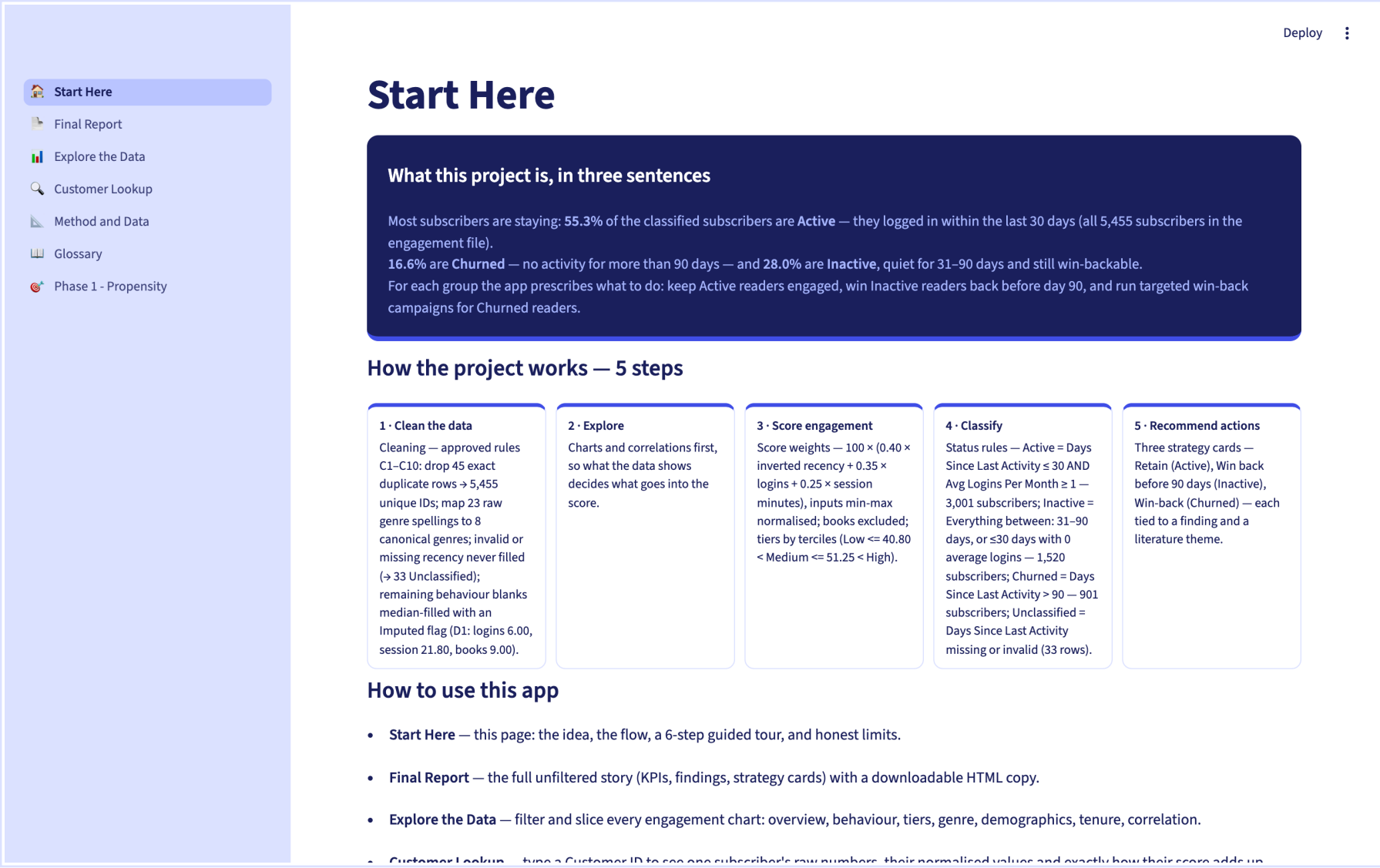
The dashboard — a guided tour

7 pages: Start Here - Final Report - Explore the Data - Customer Lookup - Method and Data - Glossary - Propensity Model

Every chart explains what it shows, how to read it, and what it means for the business — and every technical term has a plain-language definition one click away.

Page 1 · Start Here

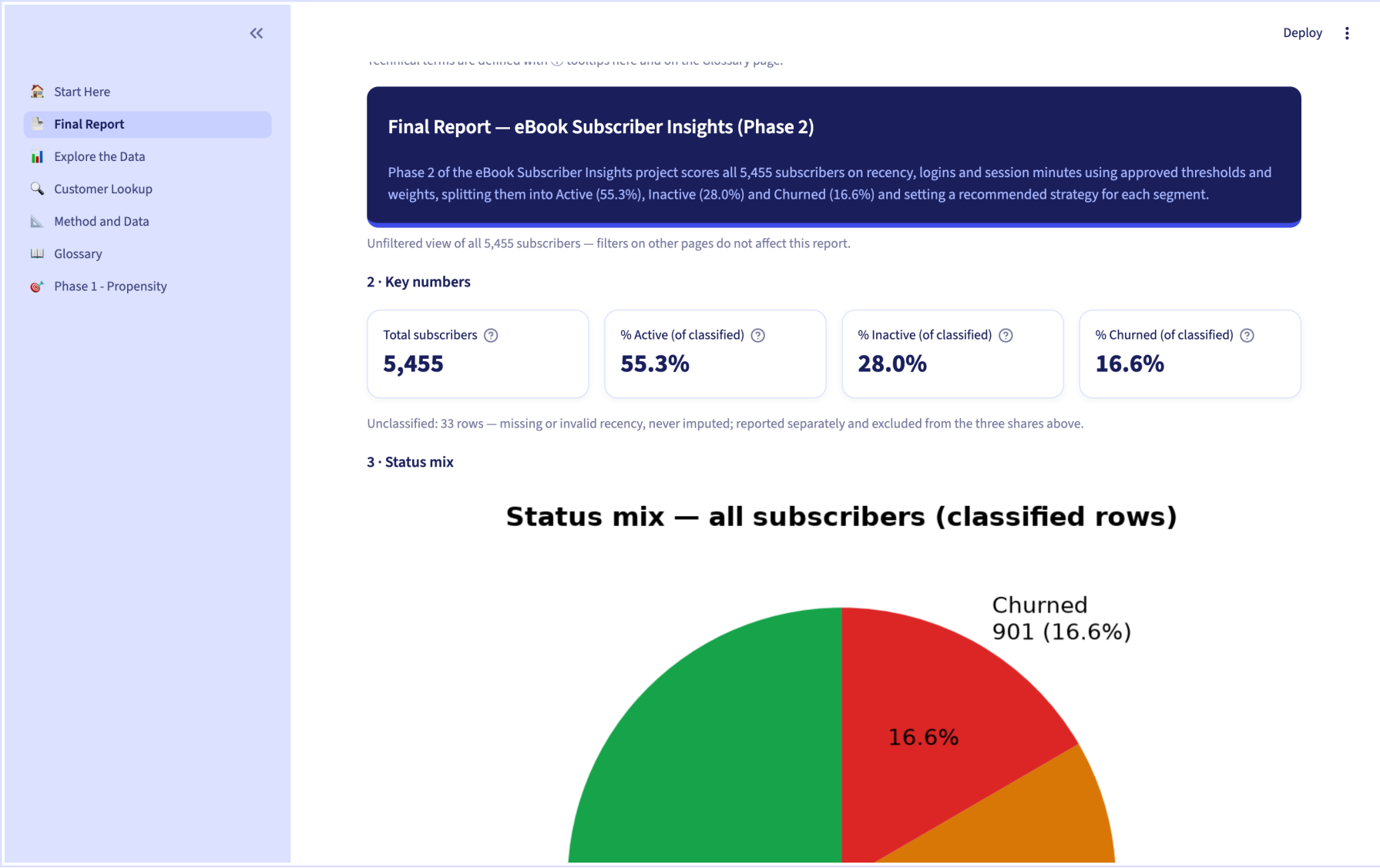
Orientation page — what the project is, in plain language



Takeaway: New visitors are told what the app can and cannot tell them before any chart is shown.

Page 2 · Final Report — key numbers

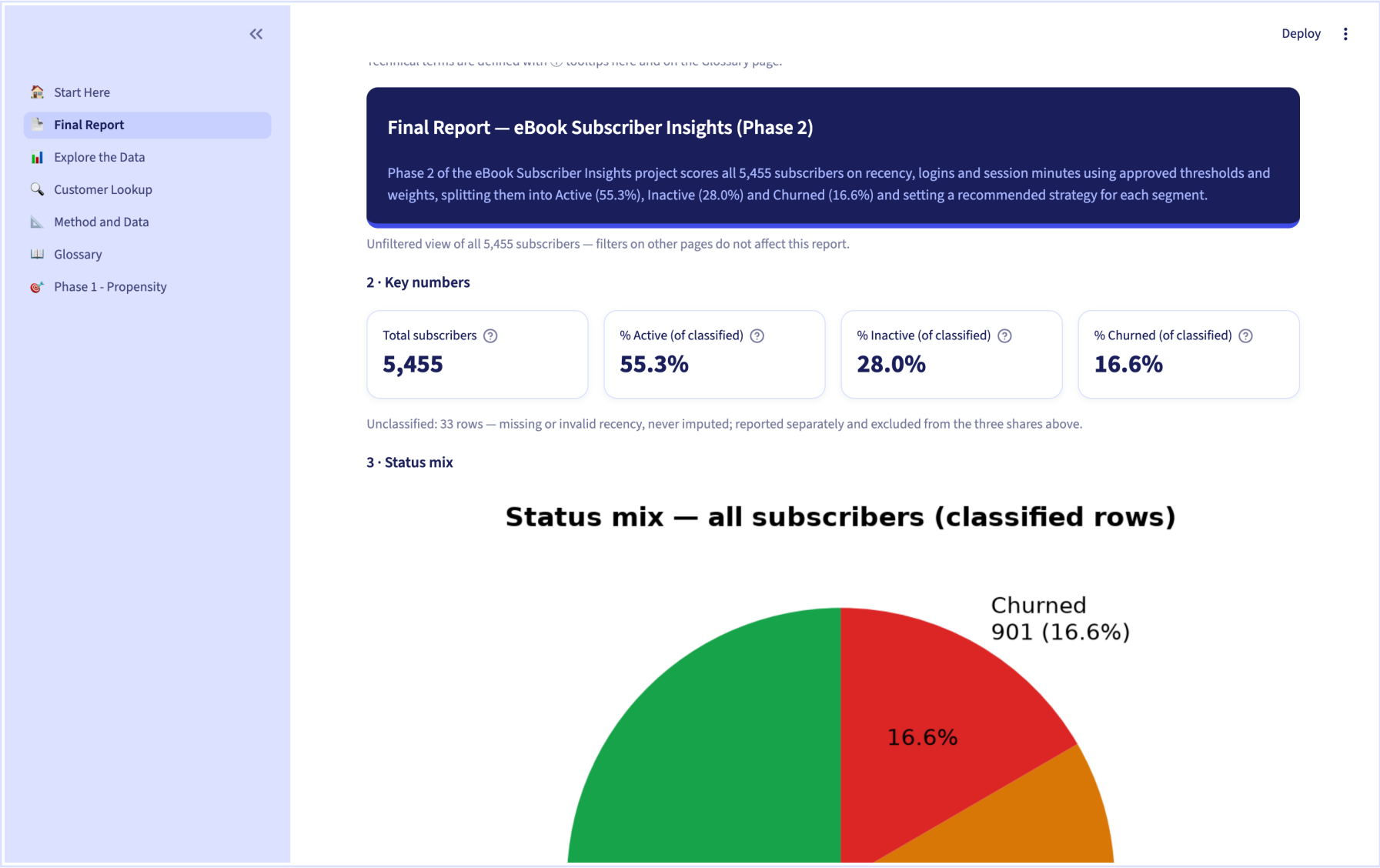
The unfiltered story of all 5,455 subscribers



Takeaway: 55.3% Active - 28.0% Inactive - 16.6% Churned — the headline every stakeholder sees first.

Page 2 · Final Report — status mix

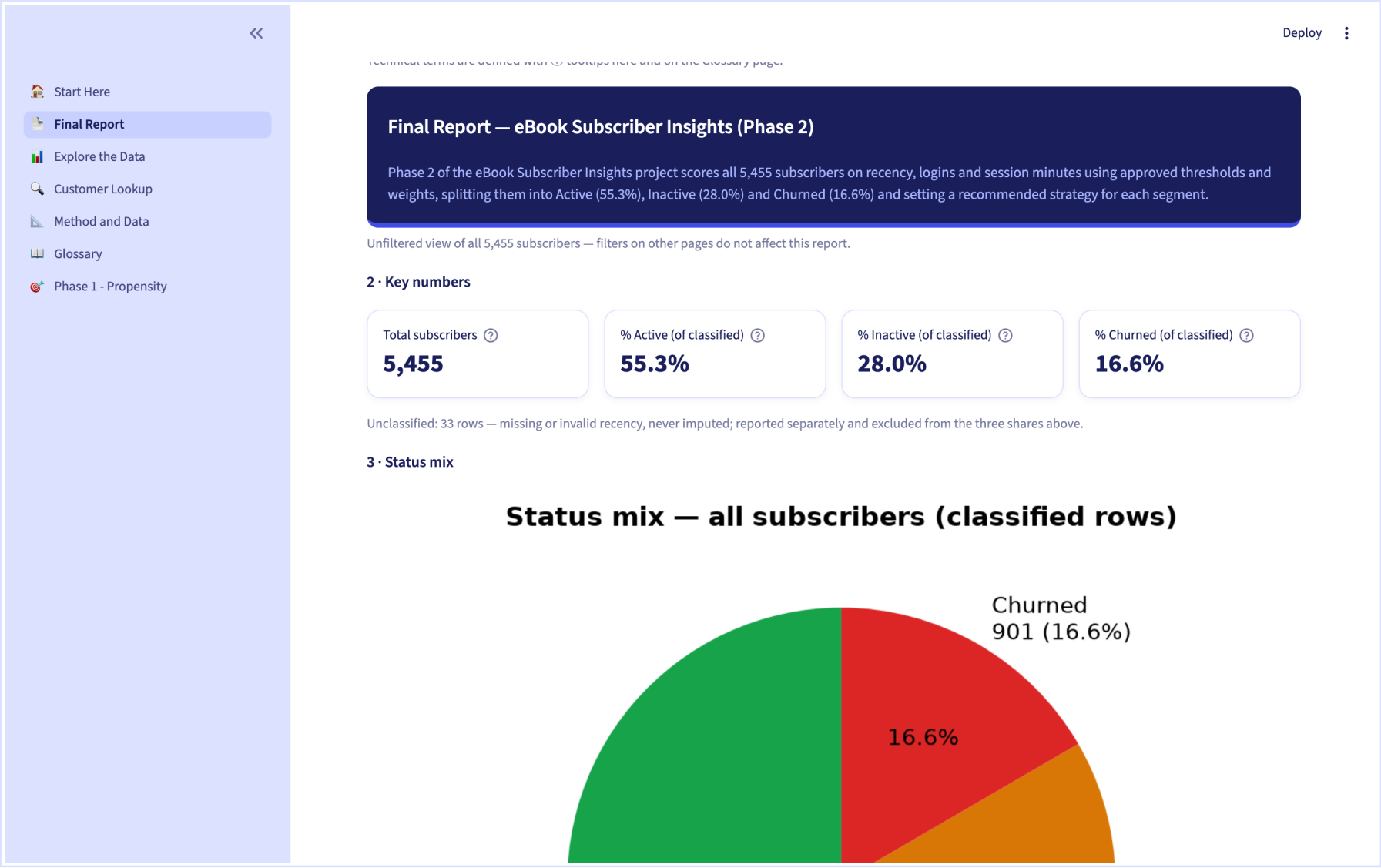
The staying-versus-leaving problem, sized



Takeaway: Most subscribers stay; the actionable middle group is the 1,520 Inactive readers.

Page 2 · Final Report — tiers & watchlist

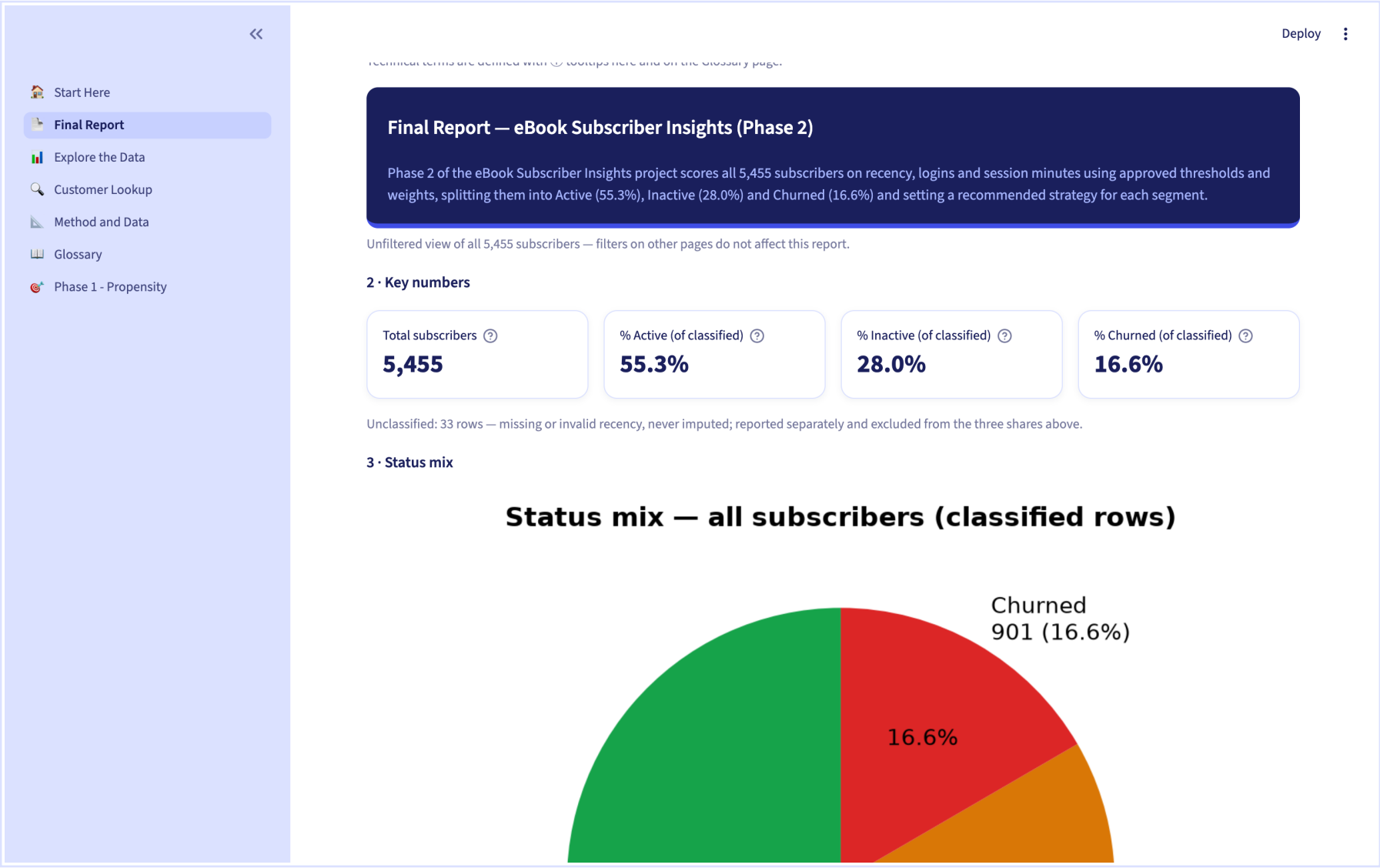
Low / Medium / High tiers, with an early-warning list



Takeaway: 11 subscribers are Active today but already Low tier — the earliest possible warning list.

Page 2 · Final Report — strategy cards

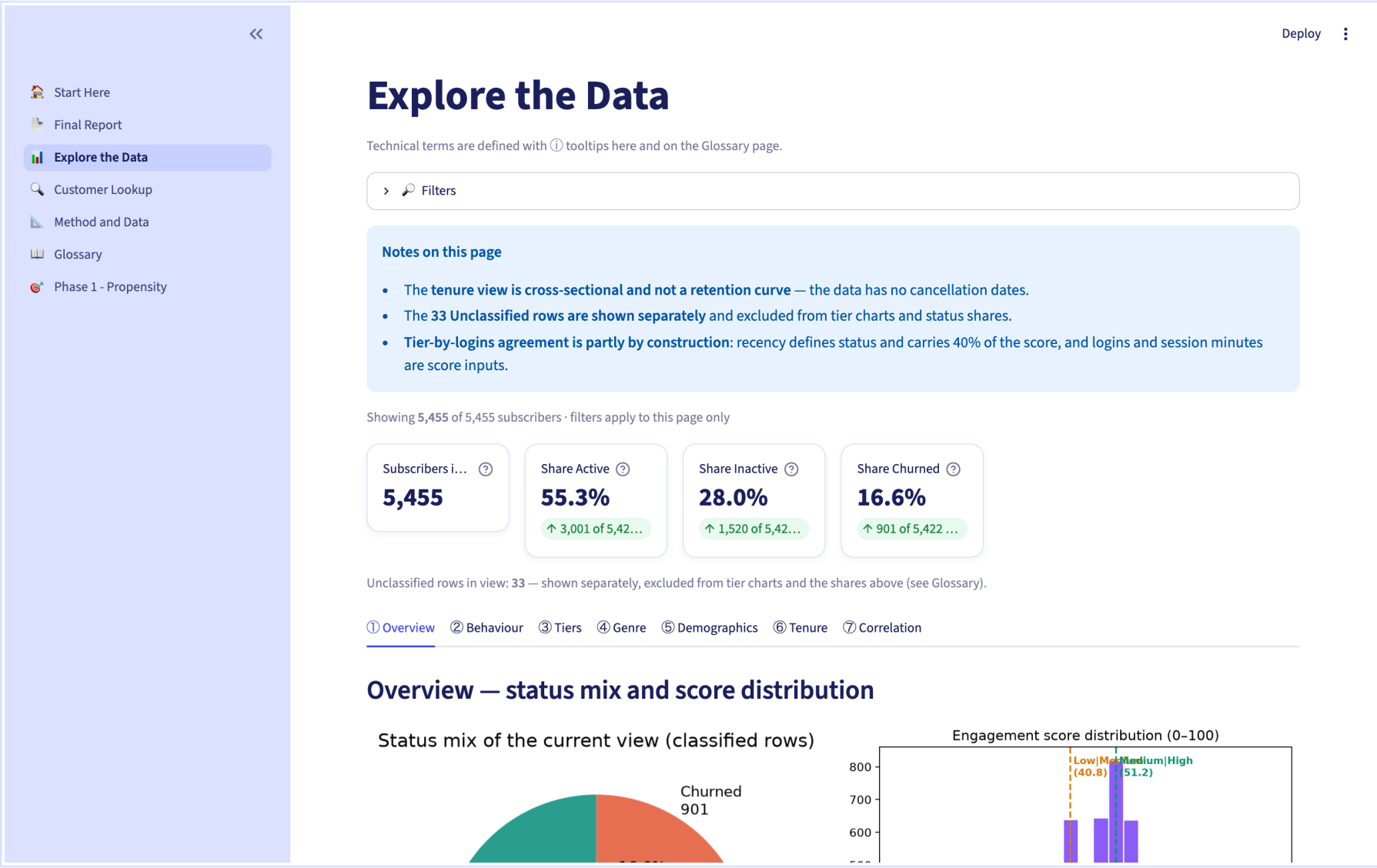
One recommended strategy per status group



Takeaway: Retain Active - Win back Inactive before day 90 - Win-back Churned — each tied to evidence.

Page 3 · Explore the Data

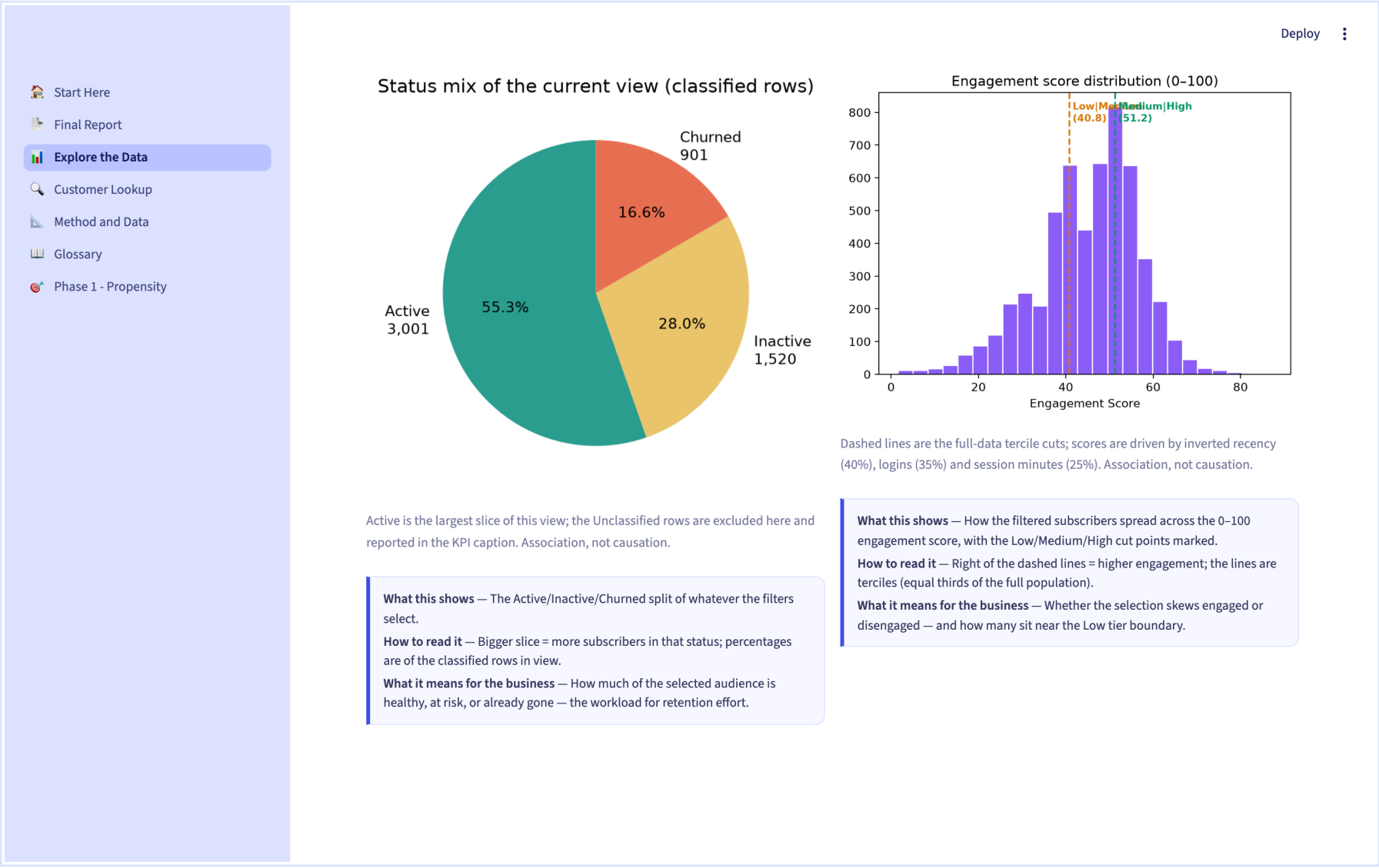
7 filterable views of the subscriber base



Takeaway: Filters update every chart below — status, tier, genre, age, occupation, education, tenure, state.

Explore · Overview

Status mix and score distribution of the current view



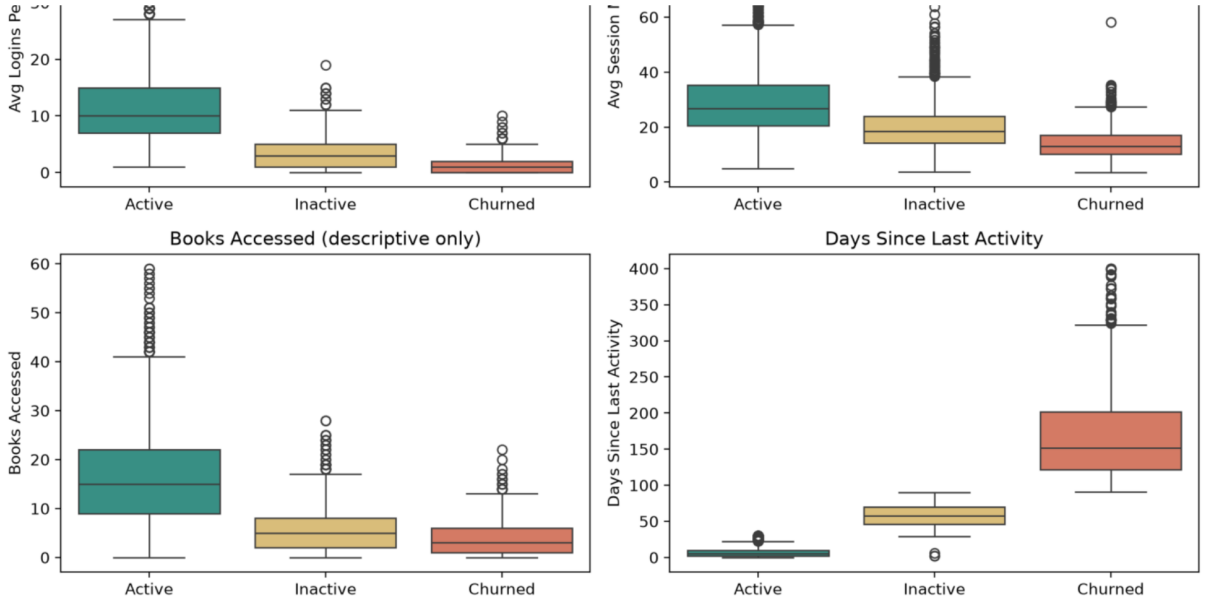
Takeaway: Where the selected audience sits between healthy, at risk and already gone.

Explore · Behaviour

Logins, session minutes, books and recency by status

Start Here
Final Report
Explore the Data
Customer Lookup
Method and Data
Glossary
Phase 1 - Propensity

Deploy



Active subscribers sit highest on logins, session minutes and books and lowest on recency; recency spreads follow the status definition itself, while logins/session are score inputs — comparisons by status are near-independent, by tier they would be circular. Association, not causation.

What this shows — The spread of logins, session minutes, books and days since activity for Active, Inactive and Churned subscribers.

How to read it — Each box shows the middle half of that group; boxes that don't overlap mean the groups genuinely differ. The recency panel follows the status rule by definition.

What it means for the business — Behaviour — not just demographics — separates the groups, so usage-based messaging (how much someone read) is the right lever.

Takeaway: Behaviour — not just demographics — separates the groups: Active 11.7 logins vs Churned 1.5.

Explore · Tiers

What the Low, Medium and High tiers really look like

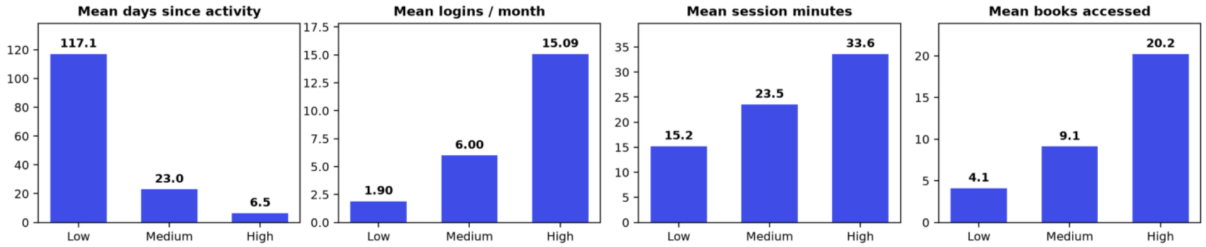
Start Here
Final Report
Explore the Data
Customer Lookup
Method and Data
Glossary
Phase 1 - Propensity

Deploy

Construction-overlap warning: status is defined on recency, so tier-vs-status agreement is partly by construction, not independent validation. Books Accessed and Subscription Tenure are NOT score inputs, so they are the independent checks.

Tier cutoffs — why terciles: Tiers use terciles of the engagement score (Low up to the lower cut (40.80), Medium up to the upper cut (51.25), High above). Reasons: fixed numbers such as 33/66 would be arbitrary; equal-sized thirds guarantee no lopsided tier; cutoffs come from the data rather than numbers we chose. Limitation: tiers are relative to this subscriber population, not absolute standards.

Mean engagement metrics by tier (classified rows — the 33 Unclassified are excluded)



books and tenure are independent of the score while recency, logins and session minutes are score inputs, so the gradient in those three is partly by construction.

What this shows — Mean recency, logins, session minutes and books in each score tier (classified rows).

How to read it — All four panels rise Low → High; books is not a score input, so its rise is the independent confirmation — the other three rise partly by construction.

What it means for the business — Tier labels track real behaviour differences, with one honest outside check (books) behind them.

Watchlist — early-warning group: 11 Active subscribers sit in the Low tier (mean recency 29.1 days, 1.45 logins, 11.3 session minutes, 4.4 books) — still Active today, engagement already Low.

› Technical details — watchlist rows and status x tier table

Takeaway: Books rise 4.1 -> 9.1 -> 20.2 across tiers — the independent check that the score means something.

Explore · Genre

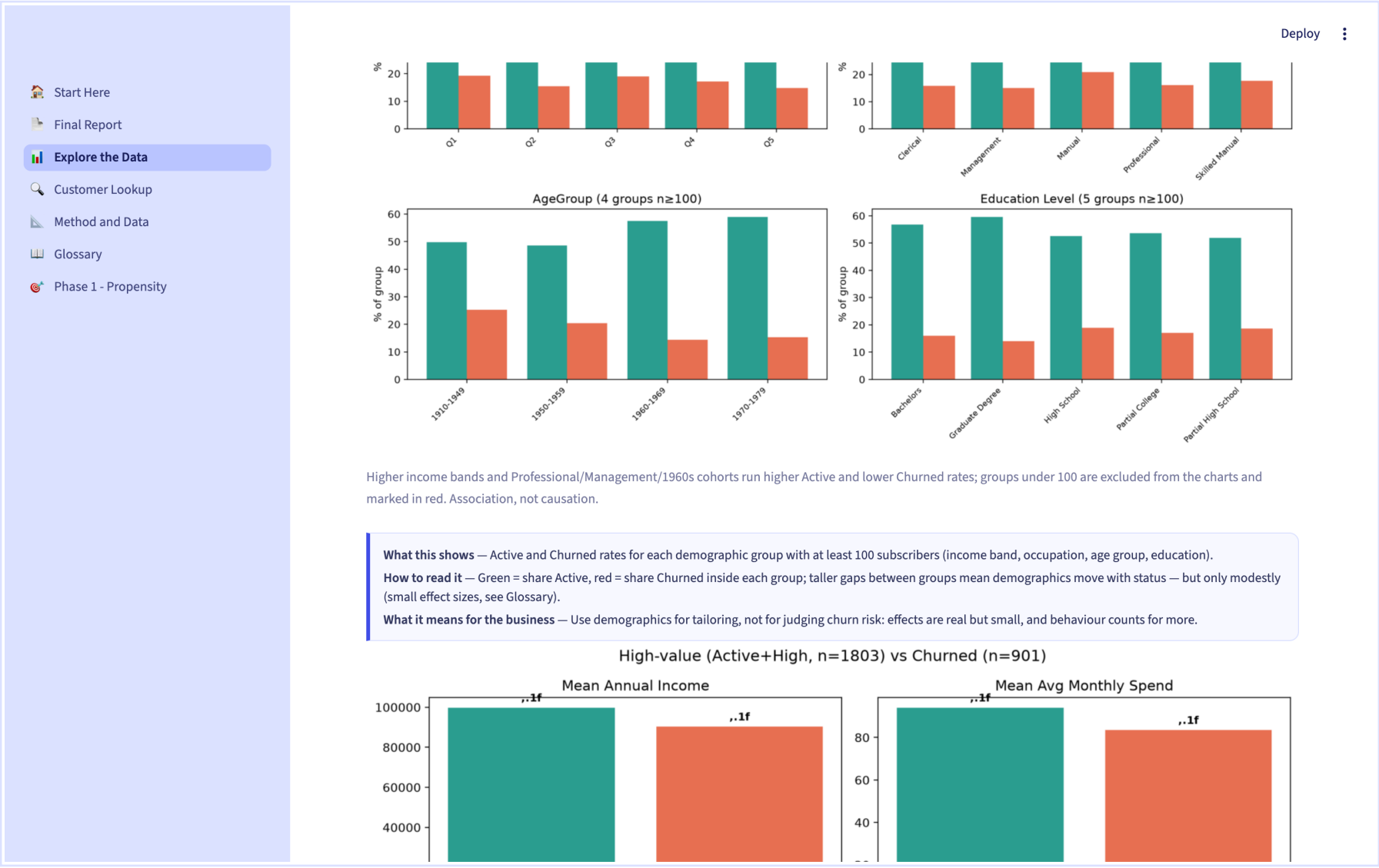
Does what people read predict whether they stay?



Takeaway: No — Active rates span only 52-58% across genres. A null finding, reported honestly.

Explore · Demographics

Income, occupation, age and education vs status



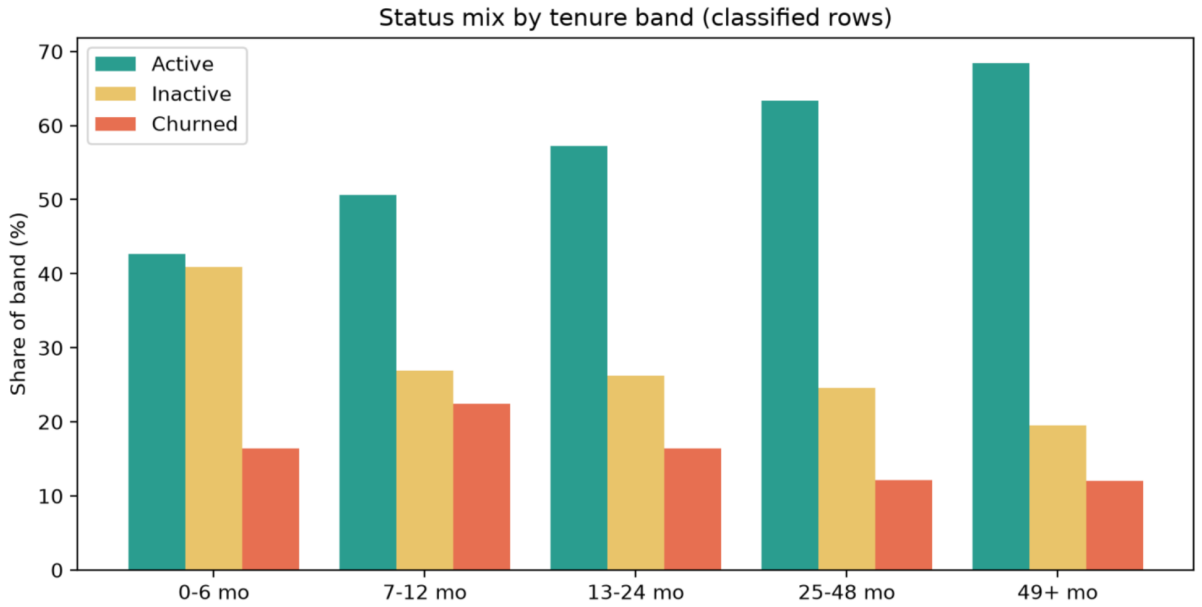
Takeaway: Demographic effects are real but small (d ~ 0.2-0.3) — tailor with them, don't judge churn on them.

Explore · Tenure

Status mix by months since subscribing

Deploy

- Start Here
- Final Report
- Explore the Data
- Customer Lookup
- Method and Data
- Glossary
- Phase 1 - Propensity



Status mix is broadly similar across tenure bands — long-tenure subscribers are not automatically more Active; cross-sectional, NOT a retention curve. Association, not causation.

What this shows — How Active/Inactive/Churned shares look inside each tenure band (months since subscribing).

How to read it — Compare the colour proportions across the five bands; similar mixes mean tenure doesn't separate statuses (churned rates are non-monotone).

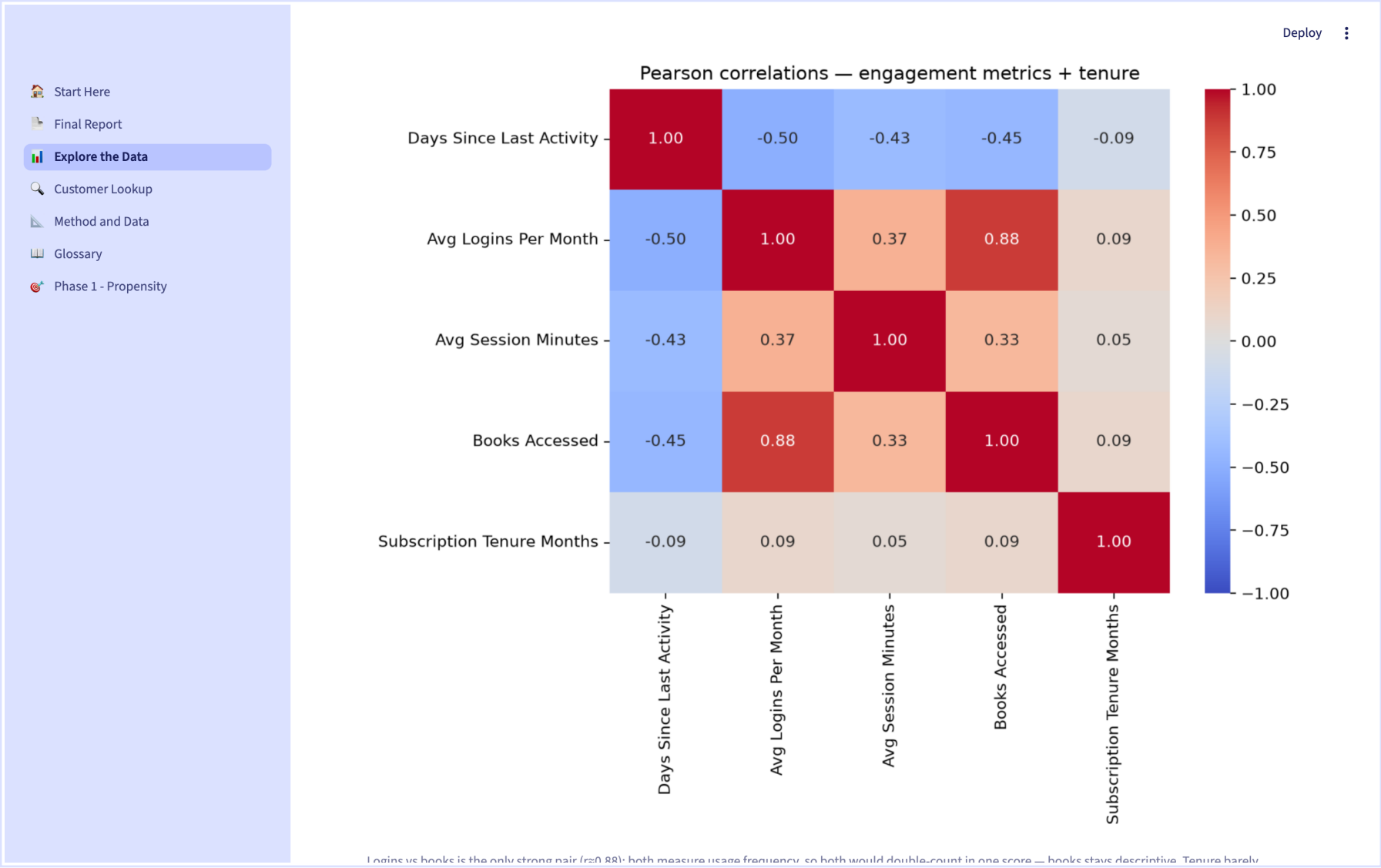
What it means for the business — Tenure alone shouldn't drive win-back targeting — and no retention-over-time claim can be made from this snapshot.

› Technical details — counts by tenure band

Takeaway: Tenure does not reliably separate churn — cross-sectional, not a retention curve.

Explore · Correlation

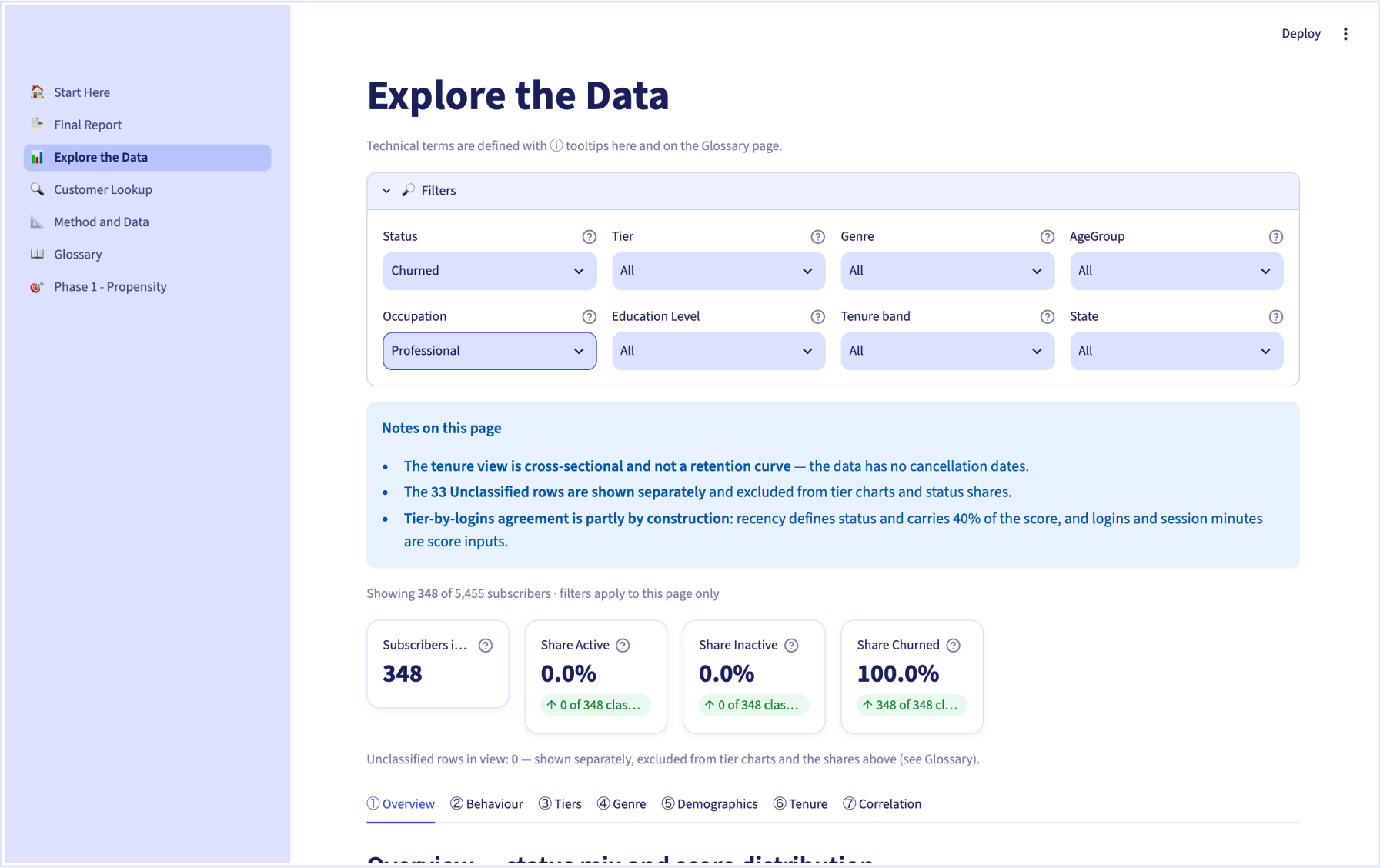
How the engagement metrics move together



Takeaway: Logins vs books r = 0.88 — the double-counting that justified excluding books from the score.

Demo · Filtering

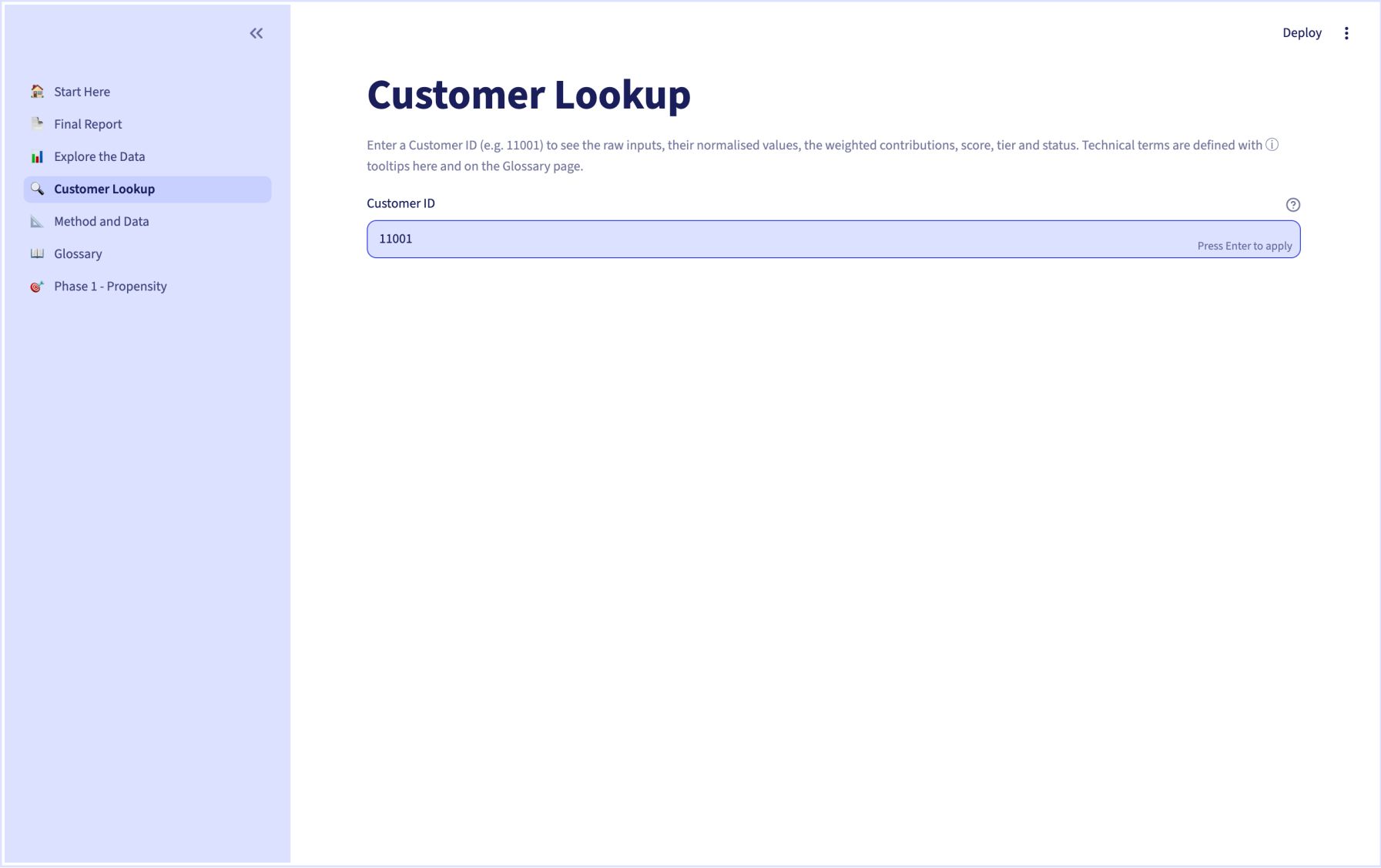
Status = Churned + Occupation = Professional



Takeaway: Every chart and KPI updates together — segment-specific strategies are one click away.

Page 4 · Customer Lookup

One subscriber, fully explained



Takeaway: Raw values -> normalised 0-1 -> weighted contributions -> score: the whole model in one row.

Page 5 · Method and Data

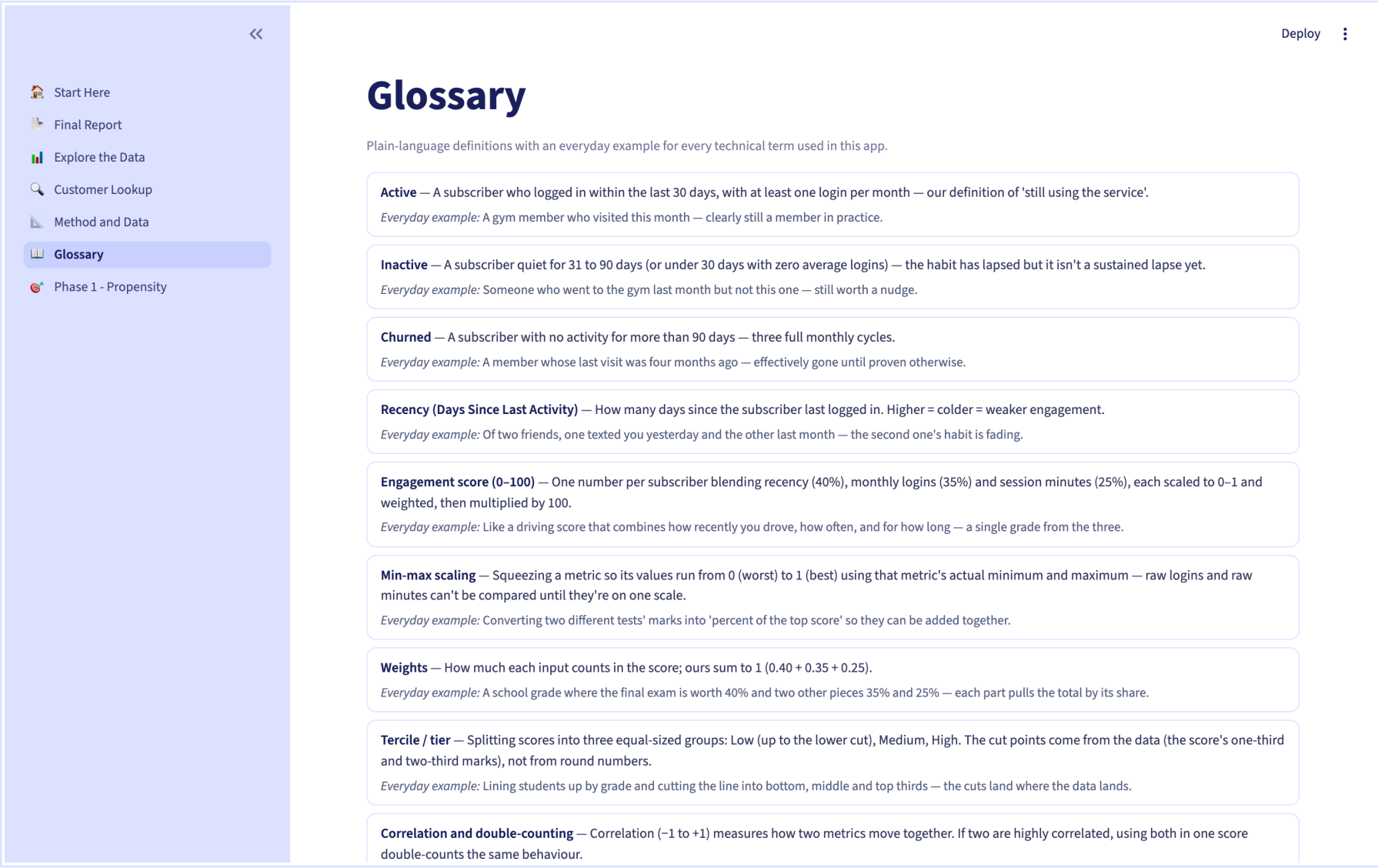
Everything needed to reproduce the analysis



Takeaway: Data profile, cleaning rules, weights and thresholds — the anti-black-box page.

Page 6 · Glossary

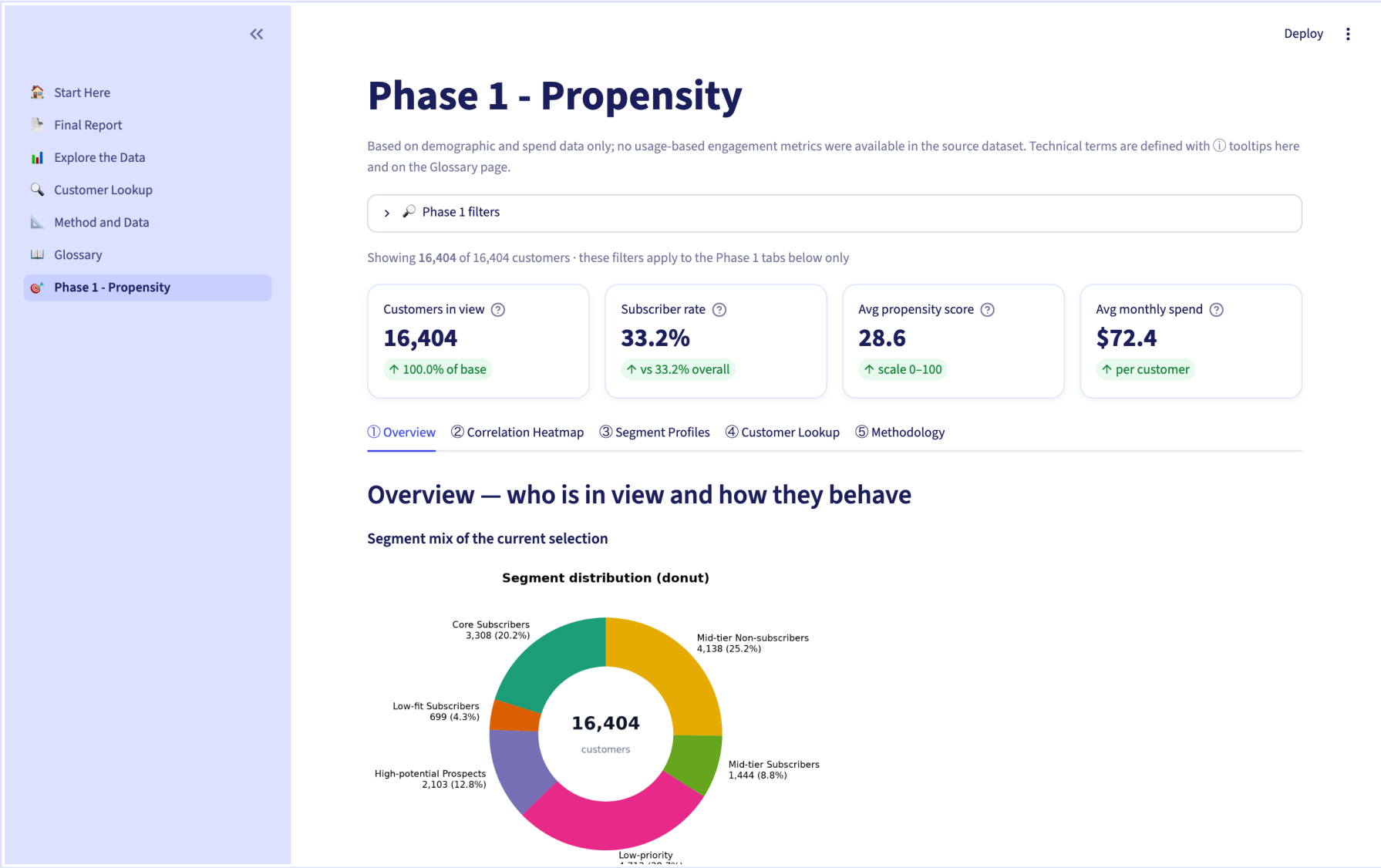
Every technical term in plain language



Takeaway: Recency, tercile, correlation, p-value, effect size — each with an everyday example.

Page 7 · Propensity Model

The original customer-file model: who is likely to subscribe



Takeaway: The first dataset's view — demographic propensity scoring and six customer segments.

What the data says

Four findings that survive the honest checks

The behaviour gap is large

Active subscribers average 11.69 logins and 29.1 session minutes a month; Churned average 1.54 and 14.2 — a 7.6x gap in visits. Median last activity: 5 days vs 152 days.

The score passes an independent test

Books and tenure are not score inputs, yet books rise 4.1 -> 9.1 -> 20.2 across tiers — the score tracks real behaviour, not just its own ingredients.

Demographics help, but only a little

High-value subscribers earn ~$99.8k vs $90.3k and spend $94.1 vs $83.5 a month — significant, but small effects (d ~ 0.2-0.3).

Genre and tenure do not predict churn

Active rates span only 52-58% across genres and churn rates are non-monotone across tenure bands — null findings reported as such.

All associations — strongest in this dataset, not proven causes. 9 tests run; Bonferroni-corrected bar 0.0056; the Fantasy-genre gap does not survive it.

What to do about it

Three strategies — each tied to a finding and a literature theme

Retain

3,001 Active - 55.3%

Keep the habit going: personalised recommendations built on each reader's own genre and history — the high-value group (1,803) is the priority.

Win back before day 90

1,520 Inactive - 28.0%

Trigger a message the day a reader passes 30 days of silence (early warning signs), send 'pick up where you left off' nudges — they still average 3.3 logins.

Win-back

901 Churned - 16.6%

Sequences keyed to declining login frequency; all 901 sit in the Low tier — intervene before day 90 for anyone sliding toward it.

Themes from the literature: declining login frequency - early warning signs - personalised recommendations - relevant communication - segment-specific strategies.

What this analysis does not claim

- Association, not causation — nothing here shows that acting on a factor will change status.
- No cancellation reasons — we show who leaves and when, never why.
- No cancellation dates — the tenure view is a cross-section, not a retention curve.
- Status is recency-based by definition; behavioural comparisons account for that.
- Only the single most-read genre per subscriber is recorded.
- The dataset is synthetic — conclusions illustrate method, not market claims.
- 33 rows with missing recency stay Unclassified — never imputed, shown separately.

Thank you · Questions welcome

Live dashboard - final_report.html (offline) - methodology, watchlist and all charts in outputs/v2/